Piter Z. Garcia Bautista

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Research Profile

Graduate researcher focused on machine learning under uncertainty, noisy observations, and limited-signal environments. Background spans reproducible ML evaluation pipelines, bandit and reinforcement-learning-based decision-making systems, quantum computing research, and fault-tolerant algorithm design. Current Graduate Assistant work (Quantum and AI Research, RIT) emphasizes reliability-first evaluation methodology, structured experimental design, and principled approaches to settings where standard supervised assumptions break down. Particularly interested in how statistical learning can be made robust and trustworthy in noisy physical and computational systems, including quantum circuits.

Research Interests

Reliable machine learning under noise and uncertainty; fault-tolerant algorithm evaluation; quantum computing and quantum-classical hybrid methods; bandit learning and sequential decision-making; reproducible benchmarking and structured experimental design.

Education

Rochester Institute of Technology — Expected Aug 2026
Master of Science, Data Science — GPA 3.9+ — *Rochester, NY*

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Data-Driven Knowledge Discovery, Neural Networks & Deep Learning, Applied Data Science, Bioinformatics Algorithms, Software Engineering for Data Science.

University of Rochester, Warner School of Education — In progress
Graduate study, Teaching Computer Science K–12 (M.S. track) — *Rochester, NY*

- NSF Robert Noyce Scholar ($30,000 full funding); Advanced Certificate in Leadership in Disability and Inclusive Practices.

Rochester Institute of Technology — 2015
Master of Science, Computer Science — *Rochester, NY*

- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics.
- MS Thesis: ASL Recognition Application — image processing and pattern recognition under partial-observation constraints.

Farmingdale State College — 2012
Bachelor of Science, Computer Engineering — *Farmingdale, NY*

- Dual degree: Electrical Engineering & Computer Engineering Technology.
- Honors: CSTEP Award & Merit; Dual BS Scholarship.

Research and Professional Experience

Rochester Institute of Technology, Software Engineering Department — Fall 2025 – Present
Graduate Assistant, Quantum and AI Research — *Rochester, NY*

- Build reproducible ML evaluation pipelines across local, Colab, and GCP environments for quantum-adjacent algorithmic experiments, with structured logging and failure-mode tracking.
- Develop bandit and decision-making systems designed for uncertain, heterogeneous, and partially observed settings, with emphasis on reliability, repeatability, and evaluation rigor.
- Co-author manuscript-level technical analyses on quantum path optimization and fault-tolerant routing under noise constraints; currently under submission.
- Design evaluation frameworks that isolate algorithmic reliability from environmental noise, enabling clean comparison between quantum-classical hybrid approaches.

Independent Research 2025 – Present
Quantum Fault-Tolerant Routing and Path Optimization *Rochester, NY*

- Developed and validated quantum-classical hybrid ML algorithms for fault-tolerant routing, modeling noise channels and decoherence effects as structured uncertainty in the learning objective.
- Produced reproducible benchmarking results comparing quantum-advantage regimes against classical baselines with rigorous statistical validation.
- Applied structured bandit frameworks (UCB, neural-UCB variants) to path selection under adversarial and stochastic noise models relevant to NISQ-era devices.

Rochester Institute of Technology 2025 – 2026
Computational Biology Research (BIO614 and BIO550) *Rochester, NY*

- Applied probabilistic and algorithmic methods to RNA structure prediction, including uncertainty-aware evaluation of competing structural hypotheses.
- Developed reproducible RNA-seq differential-expression pipelines with quality-control validation and statistical-significance reporting.

VEDADATA Inc. Sep 2022 – Aug 2024
Data Solutions Engineer *New York, NY*

- Designed scalable Python workflows for ingestion, validation, and statistical diagnostics, emphasizing data reliability and structured quality checks.
- Strengthened production reliability through anomaly detection, schema validation, and structured logging.

VIOME Inc. Jan 2021 – Sep 2022
AI Data Solutions Engineer *New York, NY*

- Developed healthcare ML workflows for feature engineering, model experimentation, and reliability-aware evaluation in AWS-based production environments.
- Implemented uncertainty-quantification and model-monitoring components to flag prediction instability under distribution shift.

Selected Technical Writing

- *Quantum Path Optimization with Fault-Tolerant Routing* — manuscript on quantum-classical hybrid ML under noise constraints and decoherence, with structured evaluation (submitted 2025–2026).
- *Fairness-Aware Bandits for Network Routing in Quantum and Clinical Settings* (DSCI601 project, manuscript-style report) — bandit learning framework applied to reliability-critical routing problems.

- *BIO614-FinalProjectProposal* (Overleaf) — formal manuscript-style computational analysis with reproducible benchmarking and structured evaluation.

Awards & Recognition

- NSF Robert Noyce Teacher Scholarship (2024–2026) — full funding for M.S. in Teaching Computer Science K–12.
- MS International Scholarship (2012–2015) — MESCYT & RIT, awarded for academic excellence.
- Honorable Mention Technology Award (2012) — NYS STEP/CSTEP, 2nd place capstone project.
- IEEE Alumni Member (2009–Present) — Member No. 90613000.

Technical Competencies

Programming: Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost; bandit/RL algorithms (UCB, neural-UCB, contextual bandits); uncertainty quantification; model evaluation
Quantum Computing: Qiskit, quantum circuit simulation, noise channel modeling, NISQ-era algorithm design
Data and Statistics: pandas, NumPy, SciPy, statistical modeling, reproducible benchmarking, structured experimental design
Infrastructure: Git/GitHub, Docker, Linux, GCP, AWS, LaTeX

Selected Fit for KTH ML for Reliable Quantum Computing Position

- Direct research experience combining ML methods with quantum computing, including fault-tolerant routing and noise-aware algorithm design in a manuscript currently under submission.
- Reproducible evaluation methodology built for settings where noise, limited observations, and hardware constraints dominate — exactly the reliability challenges in NISQ-era quantum ML.
- Bandit and sequential decision-making background provides a natural bridge to online learning and adaptive ML strategies needed in quantum-classical hybrid settings.
- Strong motivation to pursue rigorous, reliability-first ML research where careful evaluation and statistical validity are as important as model performance.